JIE WANG

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of production backend and API experience, specializing in application security, authentication/authorization, automated security testing, and incident response. Built and secured partner-facing APIs using mTLS/X.509, OIDC/JWT/JWKS, Azure, and event-driven architectures.

TECHNICAL SKILLS

Security: mTLS, X.509 certificates, certificate thumbprint validation, OIDC, JWT, JWKS, authentication/authorization, API security
Languages: C#, Python, Java, JavaScript, SQL, Scala
Cloud & Engineering: .NET, Azure DevOps, Azure API Management, Terraform, Event Hub, Key Vault, Managed Identity, Kubernetes, Docker, SpecFlow, Linux, Git
Patent: Wireless Charger Patent (China)

ENGINEERING EXPERIENCE

Software Engineer | Allegion (Schlage) | Remote, CO

Jun. 2022 – Present

APPLICATION SECURITY

• Led authentication and endpoint-authorization hardening across 4+ production services by implementing mTLS certificate-thumbprint validation, centralizing enforcement in a reusable component, and extending certificate-based authorization to sensitive endpoints and partner integrations.

• Migrated JWT validation from static keys to per-client OIDC issuer/JWKS-based trust, adding issuer-mismatch checks and negative-path coverage to support safe signing-key rotation and prevent token confusion/substitution.

• Owned production authentication failures end-to-end, resolving Google/Apple Wallet JWKS mismatches, partner certificate/thumbprint failures, and an over-broad mTLS policy exclusion while preserving enforcement controls.

BACKEND & API ENGINEERING

• Designed and shipped 10+ production APIs end-to-end across credential, wallet, metadata, and partner workflows, owning request/response contracts, service logic, response models, and unit/SpecFlow coverage.

• Owned the ProductMetadata/Balance domain across service logic, ProductMetadataChanged event publishing, and read-model maintenance, keeping downstream consumers synchronized with product-balance changes.

• Built automated unit, SpecFlow, and negative-path tests for security-sensitive API and authentication flows, covering invalid certificates, issuer mismatches, malformed tokens, and unauthorized request paths before release.

CLOUD INFRASTRUCTURE & DEVOPS

• Maintained Azure Pipelines and Terraform across dev/QA/prod for APIM, Event Hub, private endpoints, Key Vault, and network rules; replaced a legacy database-migration workflow with DbUp using Azure Managed Identity.

EVENT-DRIVEN SYSTEMS

• Designed the Azure Event Hub pipeline and shared Google/Apple lifecycle handlers with RBAC, consumer groups, and error-queue recovery, then delivered additional Cbord and PassManager event consumers.

PRODUCTION RELIABILITY & DEBUGGING

• Led application-level incident response during regular on-call rotations, resolving expired-token failures, certificate mismatches, malformed partner inputs, CI blockers, and environment misconfiguration; created runbooks for certificate failures and PACS outages to standardize diagnosis and recovery..

• Mentored summer interns through onboarding and recurring 1:1s while supporting distributed engineering teams with CI/CD walkthroughs, environment documentation, and production troubleshooting.

Software Engineer, Part-time | NeurIPS | Remote, NV

Jun. 2022 – Aug. 2022

• Refactored and modularized a ~10,000-line core codebase supporting NeurIPS conference workflows, improving maintainability and reducing risk in future feature development.

Software Engineer Intern | Allegion | Boulder, CO

May 2021 – Aug. 2021

• Built reusable Cypress automation and refactored JavaScript integration tests into shared functions/commands, reducing duplicated test code by 32%.

Research Assistant | CU Boulder CS Department | Boulder, CO

Aug. 2019 – Nov. 2020

• Developed a BERT-based retweet-prediction classifier in Python, applying NLP/ML methods to identify high-engagement social-media content.

EDUCATION

M.S. Computer Science — University of Colorado Boulder, GPA 3.9/4.0 | Aug. 2020–May 2022
B.S. Computer Science — University of Colorado Boulder, GPA 3.5/4.0 | Jan. 2017–May 2020;
Stanford Summer Session 2018